

WENJI FANG

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RESEARCH INTERESTS

- **Multimodal Circuit Foundation Model**
- **VLSI PPA Evaluation and Optimization**
- **Hardware Formal Verification**

EDUCATION

Hong Kong University of Science and Technology 2024 – Present

Ph.D. in Electronic and Computer Engineering, advised by Prof. Zhiyao Xie

Hong Kong University of Science and Technology (Guangzhou) 2022 – 2024

M.Phil. in Microelectronics, advised by Prof. Hongce Zhang and Prof. Zhiyao Xie

Nanjing University of Aeronautics and Astronautics 2017 – 2021

B.Eng. in Electrical Engineering and Automation

WORK EXPERIENCE

NVIDIA Research Jun. 2025 – Dec. 2025

PhD Research Intern Manager: Dr. Mark (Haoxing) Ren & Mentor: Dr. Rongjian Liang

GenAI for scalable physical design optimization

Hong Kong University of Science and Technology (Guangzhou) Dec. 2021 – Jul. 2022

Research Assistant Advisor: Prof. Hongce Zhang

Formal property verification of microprocessors

Peng Cheng National Laboratory Jul. 2021 – Dec. 2021

Digital IC Physical Design Intern Mentor: Prof. Biwei Xie & Prof. Yungang Bao

Logic synthesis & physical design of an SoC and tape-out

FEATURED RESEARCH PROJECTS

Multimodal Circuit Foundation Model

Pre-training general multimodal circuit foundation model for various EDA downstream tasks:

- *GenEDA*, a circuit encoder-decoder alignment framework, bridging the gap between circuit encoders and LLM decoders, published in **ICCAD'25**.
- *NetTAG*, a netlist foundation model fusing gate textual semantics and circuit graph structure, supporting netlist-stage design quality prediction and functional reasoning tasks, published in **DAC'25**.
- *CircuitFusion*, an RTL foundation model fusing three RTL modalities, including RTL code, operator graph, and functionality summary, supporting RTL-stage design quality prediction tasks, published in **ICLR'25**.
- *CircuitEncoder*, exploring RTL-netlist cross-stage alignment, published in **ASPDAC'25**.
- *Large Circuit Model* (positional paper), published in **SCIS'24**.
- A survey of Circuit Foundation Model, in **Arxiv**

RTL-Stage PPA Evaluation for Early Optimization

Predicting post-synthesis PPA metrics for RTL designs, enabling early optimization:

- *RTL-Timer*, the first RTL-stage fine-grained timing slack prediction tool, enabling RTL-stage timing optimizations, published in **DAC'24**.
- *MasterRTL*, an RTL-stage overall PPA estimation framework based on bit-level circuit representation, supporting transferred across technology library and corners, published in **ICCAD'23&TCAD'24**.

Hardware Formal Property Verification

Towards automatic and scalable hardware verification:

- *AssertLLM*, hardware verification assertions generation using LLM, published in **ASPDAC'25**.
- *WASIM*, a word-level abstract symbolic simulator on RTL, supporting formal property verification for processors, published in **TACAS'23&TCAD'23**.

FIRST-AUTHORED PUBLICATIONS

Conference:

1. **Wenji Fang**, Jing Wang, Yao Lu, Shang Liu, and Zhiyao Xie, “GenEDA: Towards Generative Netlist Functional Reasoning via Cross-Modal Circuit Encoder-Decoder Alignment”, *IEEE/ACM International Conference on Computer Aided Design (ICCAD)*, 2025.
2. **Wenji Fang**, Wenkai Li, Shang Liu, Yao Lu, Hongce Zhang and Zhiyao Xie, “NetTAG: A Multimodal RTL-and-Layout-Aligned Netlist Foundation Model via Text-Attributed Graph”, *IEEE/ACM Design Automation Conference (DAC)*, 2025.
3. **Wenji Fang**, Shang Liu, Jing Wang, and Zhiyao Xie, “CircuitFusion: Multimodal Circuit Representation Learning for Agile Chip Design”, *International Conference on Learning Representations (ICLR)*, 2025.
4. **Wenji Fang**, Shang Liu, Hongce Zhang, and Zhiyao Xie, “A Self-Supervised, Pre-Trained, and Cross-Stage-Aligned Circuit Encoder Provides a Foundation for Various Design Tasks”, *IEEE/ACM Asian and South Pacific Design Automation Conference (ASPDAC)*, 2025.
5. Zhiyuan Yan*, **Wenji Fang***, Mengming Li, Min Li, Shang Liu, Zhiyao Xie, and Hongce Zhang, “AssertLLM: Generating Hardware Verification Assertions from Design Specifications via Multi-LLMs”, *IEEE/ACM Asian and South Pacific Design Automation Conference (ASPDAC)*, 2025.
6. Shang Liu*, Yao Lu*, **Wenji Fang***, Mengming Li, and Zhiyao Xie, “OpenLLM-RTL: Open Dataset and Benchmark for LLM-Aided Design RTL Generation (Invited)”, *IEEE/ACM International Conference on Computer Aided Design (ICCAD)*, 2024.
7. **Wenji Fang**, Shang Liu, Hongce Zhang, and Zhiyao Xie, “Annotating Slack Directly on Your Verilog: Fine-Grained RTL Timing Evaluation for Early Optimization”, *IEEE/ACM Design Automation Conference (DAC)*, 2024.
8. **Wenji Fang**, Yao Lu, Shang Liu, Qijun Zhang, Ceyu Xu, Lisa Wu Wills, Hongce Zhang, and Zhiyao Xie, “MasterRTL: A Pre-Synthesis PPA Estimation Framework for Any RTL Design”. *IEEE/ACM International Conference on Computer Aided Design (ICCAD)*, 2023.
9. **Wenji Fang**, and Hongce Zhang, “WASIM: A Word-level Abstract Symbolic Simulation Framework for Hardware Formal Verification”, *International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS)*, 2023.

Journal:

1. **Wenji Fang**, Yao Lu, Shang Liu, Qijun Zhang, Ceyu Xu, Lisa Wu Wills, Hongce Zhang, and Zhiyao Xie, “Transferable Pre-Synthesis PPA Estimation for RTL Designs With Data Augmentation Techniques”, *IEEE Transactions on Computer Aided Design of Integrated Circuits and Systems (TCAD)*, 2025.
2. **Wenji Fang**, Guangyu Hu, and Hongce Zhang, “r-map: Relating Implementation and Specification in Hardware Refinement Checking”, *IEEE Transactions on Computer Aided Design of Integrated Circuits and Systems (TCAD)*, 2023.

CO-AUTHORED PUBLICATIONS

Conference:

1. Shang Liu, Yao Lu, **Wenji Fang**, Jing Wang, Zhiyao Xie, “SynC-LLM: Generation of Large-Scale Synthetic Circuit Code with Hierarchical Language Models”, *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025.
2. Wenkai Li, Yao Lu, **Wenji Fang**, Jing Wang, Qijun Zhang, Zhiyao Xie, “ATLAS: A Self-Supervised and Cross-Stage Netlist Power Model for Fine-Grained Time-Based Layout Power Analysis”, *IEEE/ACM Design Automation Conference (DAC)*, 2025.
3. Shang Liu, Jing Wang, **Wenji Fang**, Zhiyao Xie, “SynCircuit: Automated Generation of New Synthetic RTL Circuits Can Enable Big Data in Circuits”, *IEEE/ACM Design Automation Conference (DAC)*, 2025.

4. Mengming Li, Qijun Zhang, Yichuan Gao, **Wenji Fang**, Yao Lu, Yongqing Ren, Zhiyao Xie, “Profile-Guided Temporal Prefetching”, *International Symposium on Computer Architecture (ISCA)*, 2025.
5. Shang Liu, **Wenji Fang**, Yao Lu, Qijun Zhang, Zhiyao Xie, “Towards Big Data in AI for EDA Research: Generation of New Pseudo Circuits at RTL Stage”, *IEEE/ACM Asian and South Pacific Design Automation Conference (ASPDAC)*, 2025.
6. Mengming Li, **Wenji Fang**, Qijun Zhang, Zhiyao Xie, “SpecLLM: Exploring Generation and Review of VLSI Design Specification with Large Language Model”, *International Symposium of EDA (ISED)*, 2025.
7. Lei Chen, Yiqi Chen, Zhufei Chu, **Wenji Fang**, Tsung-Yi Ho, Ru Huang, Sadaf Khan, Min Li, Xingquan Li, Yu Li, Yun Liang, Jinwei Liu, Yi Liu, Yibo Lin, Guojie Luo, Hongyang Pan, Zhengyuan Shi, Guangyu Sun, Dimitrios Tsaras, Runsheng Wang, Ziyi Wang, Xinming Wei, Zhiyao Xie, Qiang Xu, Chenhao Xue, Junchi Yan, Jun Yang, Bei Yu, Mingxuan Yuan, Evangeline F.Y. Young, Xuan Zeng, Haoyi Zhang, Zuodong Zhang, Yuxiang Zhao, Hui-Ling Zhen, Ziyang Zheng, Binwu Zhu, Keren Zhu, Sunan Zou, “Large circuit models: opportunities and challenges”, *SCIENCE CHINA Information Sciences (SCIS)*, 2024.
8. Shang Liu, **Wenji Fang**, Yao Lu, Qijun Zhang, Hongce Zhang, Zhiyao Xie, “RTLCoder: Outperforming GPT-3.5 in Design RTL Generation with Our Open-Source Dataset and Lightweight Solution”, *IEEE International Conference on LLM-Aided Design (LAD)*, 2024.
9. Ang Li, Chang Wang, Feixia Bao, **Wenji Fang**, Yuxin Liang, Rui Cheng, Shilong Pan, “An Integrated Single-Shot Spectrometer With Large Bandwidth-Resolution Ratio and Wide Operation Temperature Range”, *Photonix*, 2023.

Journal:

1. Shang Liu, **Wenji Fang**, Yao Lu, Jing Wang, Qijun Zhang, Hongce Zhang, Zhiyao Xie, “RTLCoder: Fully Open-Source and Efficient LLM-Assisted RTL Code Generation Technique”, *IEEE Transactions on Computer Aided Design of Integrated Circuits and Systems (TCAD)*, 2024.

HONORS AND AWARDS

Inaugural LAD Fellowship	2025
2nd Place Award of ACM SIGDA Student Research Competition (SRC)	2025
ICCAD Student Scholar Program	2023, 2025
IEEE CEDA Travel Grant	2025
IEEE CASS Travel Grant	2025
DAC Young Fellow	2024, 2025
LAD Best Paper Nomination	2024
Full Postgraduate Studentship, HKUST	2024-Present
3rd Place Award of IEEE CEDA-HK EDathon Contest	2023
Full Postgraduate Studentship, HKUST(GZ)	2022-2024
Infineon Technology Scholarship	2020

SERVICES

Journal Reviewer

- IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)
- ACM Transactions on Design Automation of Electronic Systems (TODAES)
- IEEE Transactions on Circuits and Systems Part II: Express Briefs (TCAS-II)

TEACHING ASSISTANT

- Introduction to Computer Organization and Design, HKUST
- Electronic Design Automation for VLSI Design, HKUST
- Logic Design Automation of Digital Systems, HKUST(GZ)